

MVLU

PRACTICAL N0: 2

AIM: Describing the Distribution of a Variable

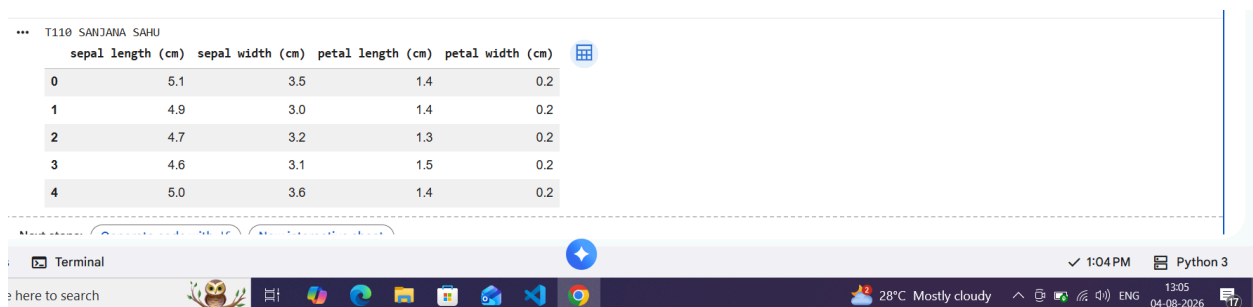
A). Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest.

CODE:

Import Libraries and Load Iris Dataset

```
print("T110 SANJANA SAHU")
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
# Load Iris Dataset
iris = load_iris()
# Convert into DataFrame
df = pd.DataFrame(iris.data, columns=iris.feature_names)
# Show first 5 rows
df.head()
```

OUTPUT:



	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	4.9	3.5	1.4
1	4.7	4.6	3.2	1.3
2	5.0	3.6	1.4	0.2
3	4.9	4.7	3.4	1.5
4	5.4	4.8	3.7	1.6

Select the Variable

```
print("T110 SANJANA SAHU")
# Select Variable
data = df['sepal length (cm)']

print(data)
```

... T110 SANJANA SAHU
0 5.1
1 4.9
2 4.7
3 4.6
4 5.0
...
145 6.7
146 6.3
147 6.5
148 6.2
149 5.9
Name: sepal length (cm), Length: 150, dtype: float64

Start coding or generate with AI.

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#Calculate Descriptive Statistics

```
print("T110 SANJANA SAHU")
print("Mean =", data.mean())
print("Median =", data.median())
print("Mode =", data.mode()[0])
print("Variance =", data.var())
print("Standard Deviation =", data.std())
print("Minimum =", data.min())
print("Maximum =", data.max())
```

... T110 SANJANA SAHU
Mean = 5.843333333333334
Median = 5.8
Mode = 5.0
Variance = 0.6856935123042505
Standard Deviation = 0.8280661279778629
Minimum = 4.3
Maximum = 7.9

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B) Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

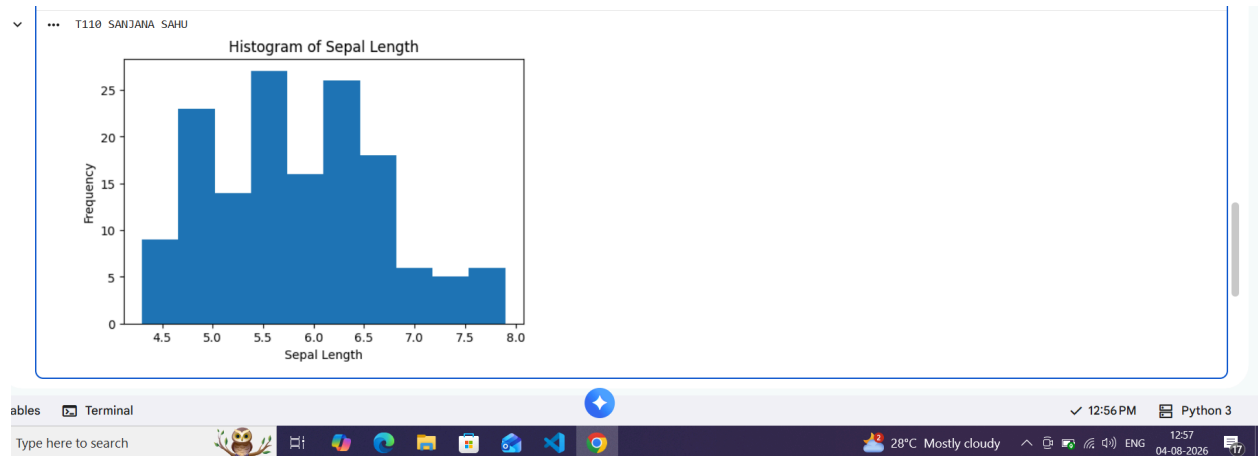
CODE:Histogram of Sepal Length

```
print("T110 SANJANA SAHU")
plt.figure(figsize=(6,4))
plt.hist(data, bins=10)

plt.title("Histogram of Sepal Length")
plt.xlabel("Sepal Length")
plt.ylabel("Frequency")
plt.show()
```

OUTPUT :

T110 TYCS SANJANA SAHU

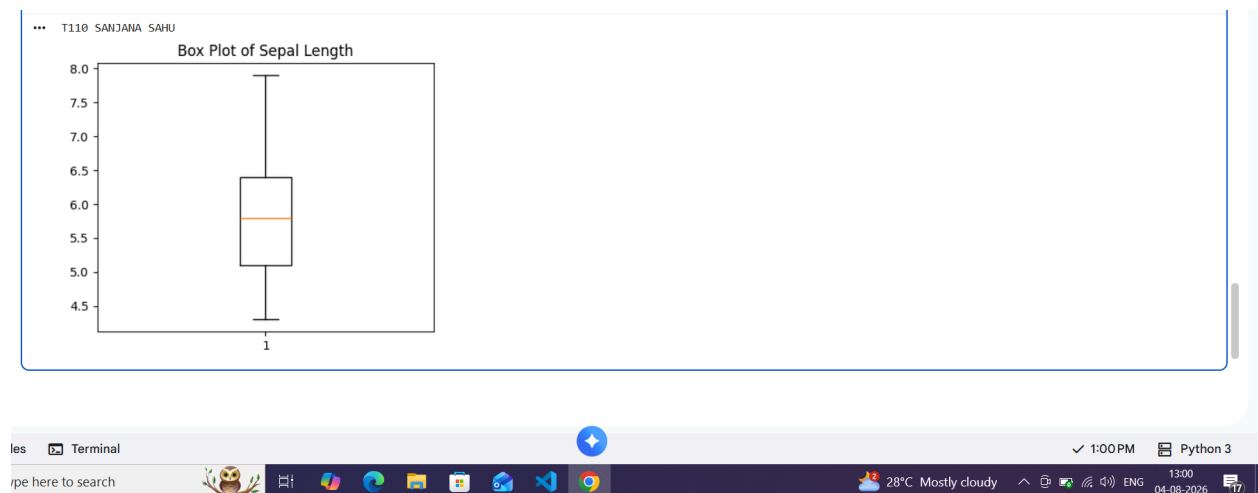


#Box Plot of Sepal Length

CODE:

```
print("T110 SANJANA SAHU")
plt.figure(figsize=(5,4))
plt.boxplot(data)
plt.title("Box Plot of Sepal Length")
plt.show()
```

OUTPUT:



Variable Used	Sepal Length (cm).
Mean	Average value.
Median	Middle value.
Mode	Most frequent value.
Variance	Measures data spread.
Standard Deviation	Measures variation from the mean.
Minimum	Smallest value.
Maximum	Largest value.
Histogram	Shows frequency distribution.
Box Plot	Shows distribution and outliers.
Pandas	Library for data handling.
Matplotlib	Library for plotting graphs.
Scikit-learn	Library providing the Iris dataset.
DataFrame	Table of rows and columns.